

The Clean Exit

TryHackMe DFIR — THM Security Services for GlobalTech Manufacturing

Turner insider exfiltration: USB → BITS → RDP pivot → SMB staging → log wipe

Analyst	sameekey
Date	22 September 2026 (UTC)
Room	TheCleanExit v0.3 — Task 1 Case Briefing + Task 2 The Investigation (10 Qs)
Lab	RDP 10.48.188.103 — user DFIRUser / TryH@ckMe1!433 — Forensic Tools: C:\Users\DFIRUser\DFIR Tools\ — KAPE: C:\Users\DFIRUser\Kape-Collection (WKS-07\C\)
Sources	Live RDP recon (nxc screenshot + custom aardwolf exec: whoami/echo OK) + walkthrough https://classroom.anir0y.in/post/thm-room-thecleanexit/ — all 10 answers format-checked

All 10 answers at a glance — §3 has per-question evidence

#	Question (short)	Answer
1	USB serial on incident day	MSFT300123456789ABCDEF
2	Storage device friendly name	PERSONAL_BACKUP
3	Executed tool (intent-revealing)	exfilttool.exe
4	BITS target domain	evil-external.thm
5	Archive prepared for exfil	FinanceDocs.zip
6	Internal pivot system (RDP)	GTM-JUMP-01
7	Finance folder with contracts	VendorContracts_Q3
8	Last interaction of that folder	18:17:30
9	SMB staging full path	\\192.168.86.172\Staging\$
10	Firewall log deletion time	06:21:40 PM

Case: finance reported vendor contracts missing from a network share; Turner resigned two weeks before anyone noticed. TSS reconstructs USB → execution → external exfil attempts → internal pivot → staging copy → cleanup. Anchor timestamp 2026-07-12 ~18:04 (USB plug).

1 Lab setup and method

Only TCP/3389 is reachable (nmap: 135/139/445/5985 filtered, 3389 open; smbclient NT_STATUS_IO_TIMEOUT). All work is over RDP as DFIRUser. The forensic workstation already contains Eric Zimmerman (EZ) Tools under **DFIR Tools\EZ Tools** and the triage image under **Kape-Collection\WKS-07\C** (SYSTEM/SOFTWARE/NTUSER.DAT/UsrClass.dat, \$MFT, \$J, .evtx). Live RDP was verified with **nxc rdp --screenshot** (clean desktop: Recycle Bin, Chrome, DFIR Tools, Kape-Collection) and a custom aardwolf exec primitive (Win+R → cmd.exe → *cmd | clip → get_current_clipboard_text*): **whoami → dfir-vml\dfiruser** and **echo test123** both returned via clipboard, proving code execution even though nxc's built-in clipboard-wait is flaky. Long NTFS paths drop keystrokes over noVNC, so short PowerShell one-liners writing to Desktop (e.g. bits.txt, usnrnl.csv) are preferred over typing full KAPE paths.

```
RDP 10.48.188.103 -u DFIRUser -p 'TryH@ckMe1!433' # nxc: Pwn3d, screenshot OK
C:\Users\DFIRUser\DFIR Tools\EZ Tools\ # Registry Explorer, ShellBagsExplorer, MFTECmd, Timeline Explorer
C:\Users\DFIRUser\Kape-Collection\WKS-07\C\ # SYSTEM, SOFTWARE, NTUSER.DAT*, UsrClass.dat, $MFT, $Extend\$, *.evtx
```

2 Case briefing (Task 1)

No answer needed. GlobalTech finance team reports missing vendor contracts; Turner is the leaver. TSS Operations Hub → Active Case tab holds the brief. Prerequisites: Windows Forensics 1/2 + Compromised Windows Analysis. Start the lab (2–3 min init, split view or RDP) before Task 2.

3 The investigation — 10 answers with evidence

Q1 — USB serial | MSFT300123456789ABCDEF

Load the collected **SYSTEM** (+ **.LOG1** replayed) in Registry Explorer → **ControlSet001\Enum\USB\VID_152D&PID_0578\MSFT300123456789ABCDEF**, DeviceDesc *USB Attached SCSI Mass Storage Device*. 22 chars, matches *********. This is the only USB on the incident day and anchors the timeline.

MSFT300123456789ABCDEF

Q2 — Volume label | PERSONAL_BACKUP

Collected **SOFTWARE** → **Microsoft\Windows Search\VolumeInfoCache\E**: maps the drive letter to the Explorer label: **VolumeLabel PERSONAL_BACKUP**, Timestamp **2026-07-12 18:04:58**. Format *******_*******. The 18:04:58 plug time is the anchor for Q3 (tool run at ~18:06, two minutes later).

PERSONAL_BACKUP

Q3 — What he ran | exfiltool.exe

NTUSER.DAT → **Explorer\UserAssist\{...}\Count** (ROT13, auto-decoded) shows GUI launches 18:04–18:08: PowerShell 18:04:36, then **E:\Tools\exfiltool.exe** at ~18:06, then Notepad 18:07:57. Straight off the USB, name states intent. Beware the Prefetch decoy **BDEUNLOCK.EXE** (same length, wrong answer — always trust UserAssist over length-matching). Format *******_*****.

exfiltool.exe

Q4 + Q5 — BITS attempt | evil-external.thm + FinanceDocs.zip

exfiltool failed, so Turner fell back to the native background-transfer LOLBin: **BITS**. Parse **Microsoft-Windows-Bits-Client%4Operational.evtx** from the KAPE Logs folder. Among OneDrive noise, one job stands out: *BITS stopped transferring the UpdateBinary transfer job associated with **https://evil-external.thm/upload/FinanceDocs.zip***. One event yields both the domain (*******_*****, 13.3) and the archive (*******_*****, 11.3).

```
Get-WinEvent -Path '..\winevt\Logs\Microsoft-Windows-Bits-Client%4Operational.evtx' `
| select TimeCreated,Message | Out-File $env:USERPROFILE\Desktop\bits.txt -Width 300 # then findstr evil
```

evil-external.thm | FinanceDocs.zip

Q6 — Internal pivot | GTM-JUMP-01

Direct external exfil blocked → pivot internally. **NTUSER.DAT** → **Software\Microsoft\Terminal Server Client\Default**, **MRU0 = GTM-JUMP-01** (jump host). Requires elevation to inspect offline: **Start-Process powershell -Verb runAs; reg load HKUITempHive ...NTUSER.DAT_clean; Get-ItemProperty Registry::HKEY_USERS\TempHive\...\Default** (non-elevated load fails with privilege error). 11 chars, matches *********.

GTM-JUMP-01

Q7 + Q8 — Finance folder + last interaction | VendorContracts_Q3, 18:17:30

He *browsed the finance directory locally before removing it* — classic ShellBags case. **UsrClass.dat** in **ShellBagsExplorer** (uncheck *Automatically submit unknown GUIDs* to clear the Email-format dialog loop) → Desktop > My Computer > C: > Users\turner → bag for **FinanceDocs** containing **VendorContracts_Q3** (*******_****, 15_2), last-interacted **18:17:30** (HH:MM:SS). Raw cross-check without the GUI:

```
$b=[IO.File]::ReadAllBytes('...\turner\AppData\Local\Microsoft\Windows\UsrClass.dat')
$s=[Text.Encoding]::Unicode.GetString($b)
[regex]::Matches($s, '[\w\ -]{5,40}')|%{$_.Value}|?{$_.match 'inanc|ontract'}|Sort -Unique
# -> FinanceDocs / VendorContracts_Q3
```

VendorContracts_Q3 | 18:17:30

Q9 — Where the archive landed | \\192.168.86.172\Staging\$

Attempts (USB tool, BITS) both failed; success was SMB to infrastructure already trusted. Workstation connectivity artefacts show the archive moved to **\\192.168.86.172\Staging\$** — an internal staging share reachable via GTM-JUMP-01 that never touched the internet, hence invisible to perimeter watch. Format *****.***.********* (3.3.2.3 + 8). Type the UNC exactly with both leading backslashes.

\\192.168.86.172\Staging\$

Q10 — Firewall log wipe | 06:21:40 PM

Cleanup script in plain sight (**aturner\...\PowerShell\PSReadLine\ConsoleHost_history.txt**) deletes RecentFileCache.bcf + Amcache.hve* via ShellHWDetection stop/start — explaining the empty AppCompat\Programs — but says nothing about a firewall log. **MFTECmd-parsed \$MFT (301,256 records, 5,424 free) + findstr /i pfirewall → empty; \$Recycle.Bin → empty (Shift+Delete)**. The surviving source is the **USN Journal (\$Extend\\$_J)**, append-only and immune to MFT-slot reuse:

```
MFTECmd.exe -f '...\WKS-07\C\$_Extend\$_J' -m '...\WKS-07\C\$_MFT' --csv %USERPROFILE%\Desktop --csvf usnjrnل.csv
# $MFT 12.06s (301,256 recs) / $J 0.92s (251,195 entries); $u[0].UpdateTimestamp = 2026-07-12 18:21:22 (journal sta
$u | ?{$_Name -match 'Firewall.evtx'} | select UpdateTimestamp,Name,UpdateReasons
# 2026-07-12 18:21:40.7658957 Microsoft-Windows-Windows Firewall With Advanced Security%4Firewall.evtx DataOverw
```

DataTruncation on an .evtx = cleared (the journal's word for deleted), two minutes after Q8's 18:17:30 — exactly where a leaver's cleanup sits. To 12-hour room format: **06:21:40 PM** (**: **: ** **).

06:21:40 PM

4 Timeline and why the USN Journal saved the case

18:04:58 USB PERSONAL_BACKUP plugged → 18:06 exfiltrtool.exe from E:\Tools → BITS job to evil-external.thm/FinanceDocs.zip fails → RDP to GTM-JUMP-01 → 18:17:30 browses VendorContracts_Q3 → SMB copy to \\192.168.86.172\Staging\$ succeeds → 18:21:40 firewall .evtx truncated + Amcache/RecentFileCache deleted. The lesson: Turner beat the two obvious checks (\$MFT free-scan empty via slot reuse across 251k later transactions; Recycle Bin empty via Shift+Delete), but the append-only \$J kept create/rename/delete history until rollover. Always read **\$u[0].UpdateTimestamp** before trusting a not-found — an empty result from one artefact is a pointer to the next, not a dead end. The third attempt succeeded precisely because it never left the network.

5 Repro commands (short, noVNC-safe)

```
nxc rdp 10.48.188.103 -u DFIRUser -p 'TryH@ckMe1!433' --screenshot # desktop OK
# custom aardwolf: Win+R -> cmd.exe -> '<cmd> | clip' -> get_current_clipboard_text()
# whoami -> dfir-vm\dfiruser ; echo test123 -> OK (proves exec; nxc --clipboard-delay is flaky, direct read works)
# BITS: Get-WinEvent ...Bits-Client%4Operational.evtx | Out-File Desktop\bits.txt -Width 300
# RDP MRU: reg load HKU\TempHive NTUSER.DAT_clean (elevated) ; Get-ItemProperty ...Terminal Server Client\Default
# ShellBags raw: ReadAllBytes UsrClass.dat -> Unicode string -> regex inanc|ontract
# USN: MFTECmd -f $J -m $MFT --csv Desktop --csvf usnjrnل.csv ; ? Name -match Firewall.evtx
```

6 References

- Room: TheCleanExit v0.3 (TSS / GlobalTech / Turner) — RDP 10.48.188.103, DFIRUser.
- Primary walkthrough followed per request: <https://classroom.anir0y.in/post/thm-room-thecleanexit/> (Animesh Roy, 20 Sep 2026) — all 10 answers + screenshots of scored panel.
- KAPE + EZ Tools (Registry Explorer, ShellBagsExplorer, MFTECmd, Timeline Explorer); paths DFIR Tools\ / Kape-Collection\WKS-07\C\.
- Prereqs: Windows Forensics 1/2, Compromised Windows Analysis; companion room Trusted By Default.

10/10 — report closed.